Practical 3

Aim : Create a simple REST service.

Step 1 : Create a database and table in Mysql

Mysql > create database prac1;
Mysql > use prac1;
Mysql > show tables;
Mysql > create table player(playername varchar(50),playerage int,country varchar(50),runs int);
Mysql > show tables;
Mysql > desc player;
Mysql > select * from player;

```
C:\Program Files\MySQL\MyS

Enter password: ****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 1
Server version: 5.1.73-community MySQL Community Server (GPL)

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> create database prac1;
Query OK, 1 row affected (0.02 sec)

mysql> use prac1;
Database changed
mysql> show tables;
Empty set (0.00 sec)

mysql> create table player(playername varchar(50),playerage int,country varchar(50),runs int);
Query OK, 0 rows affected (0.03 sec)

mysql> show tables;
+-----------------+
| Tables_in_prac1 |
+-----------------+
| player          |
+-----------------+
1 row in set (0.00 sec)

mysql> desc player;
+------------+-------------+------+-----+---------+-------+
| Field      | Type        | Null | Key | Default | Extra |
+------------+-------------+------+-----+---------+-------+
| playername | varchar(50) | YES  |     | NULL    |       |
| playerage  | int(11)     | YES  |     | NULL    |       |
| country    | varchar(50) | YES  |     | NULL    |       |
| runs       | int(11)     | YES  |     | NULL    |       |
+------------+-------------+------+-----+---------+-------+
4 rows in set (0.01 sec)

mysql> select * from player;
Empty set (0.00 sec)

mysql> |
```
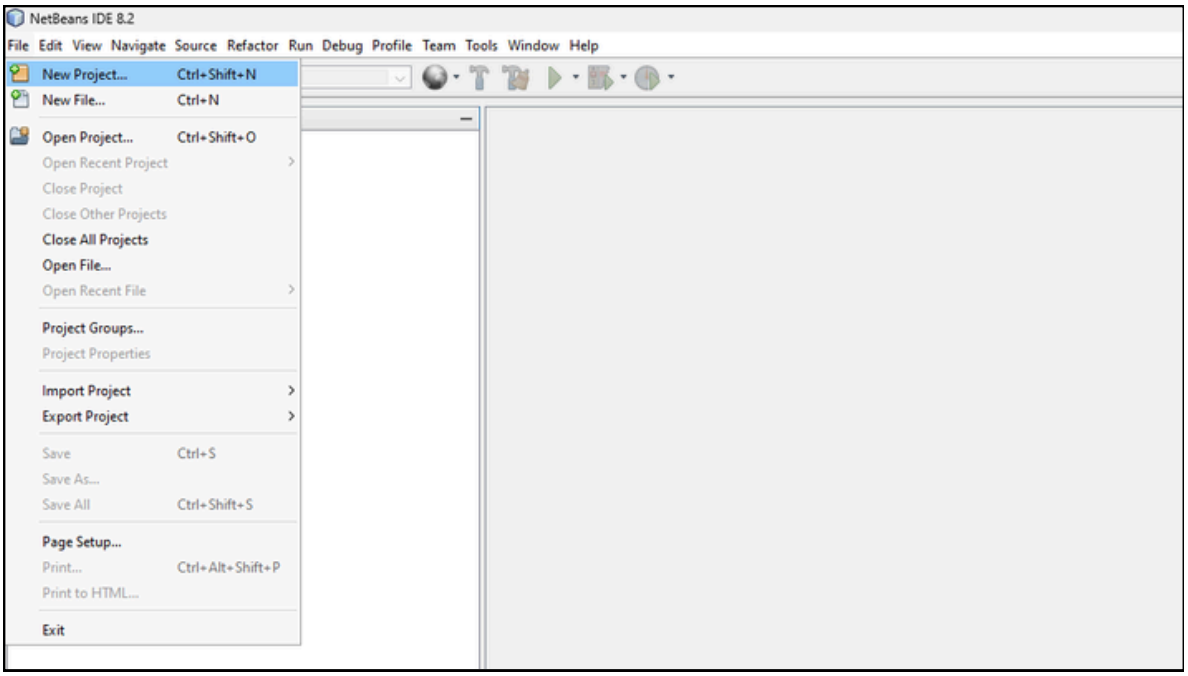
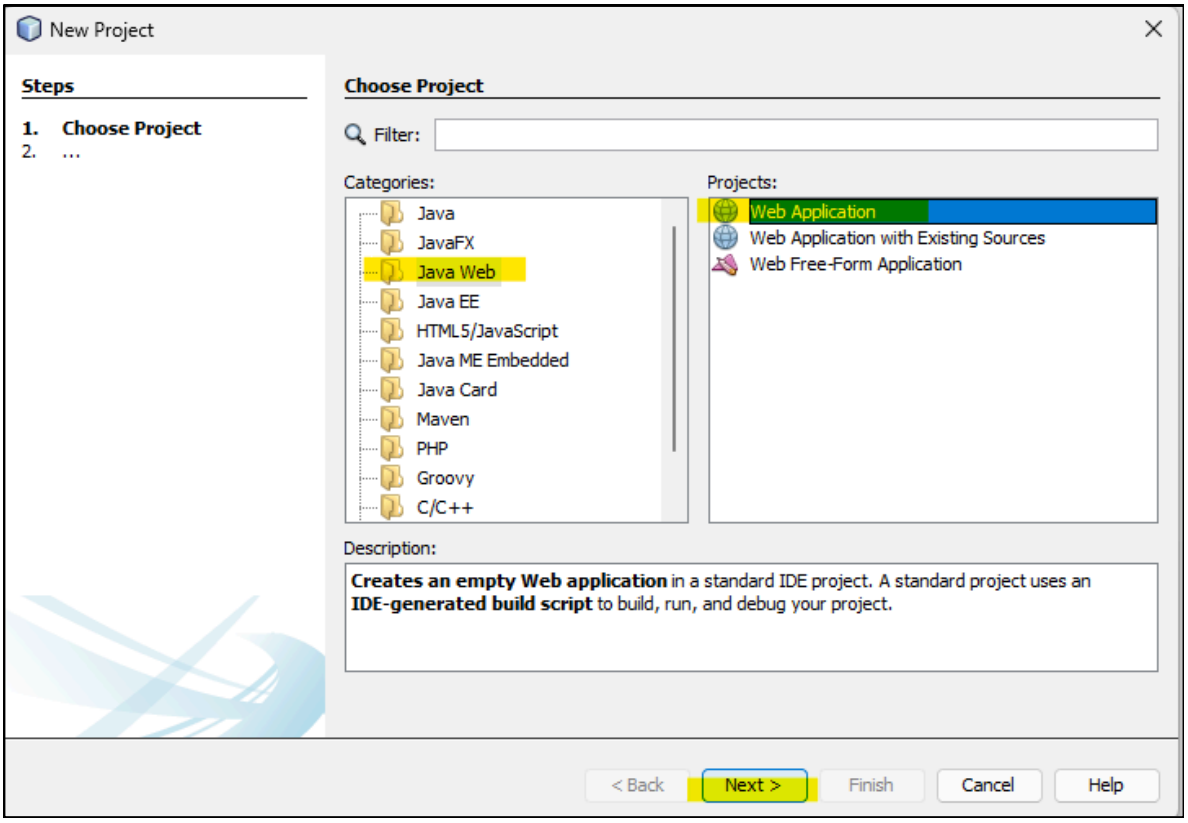
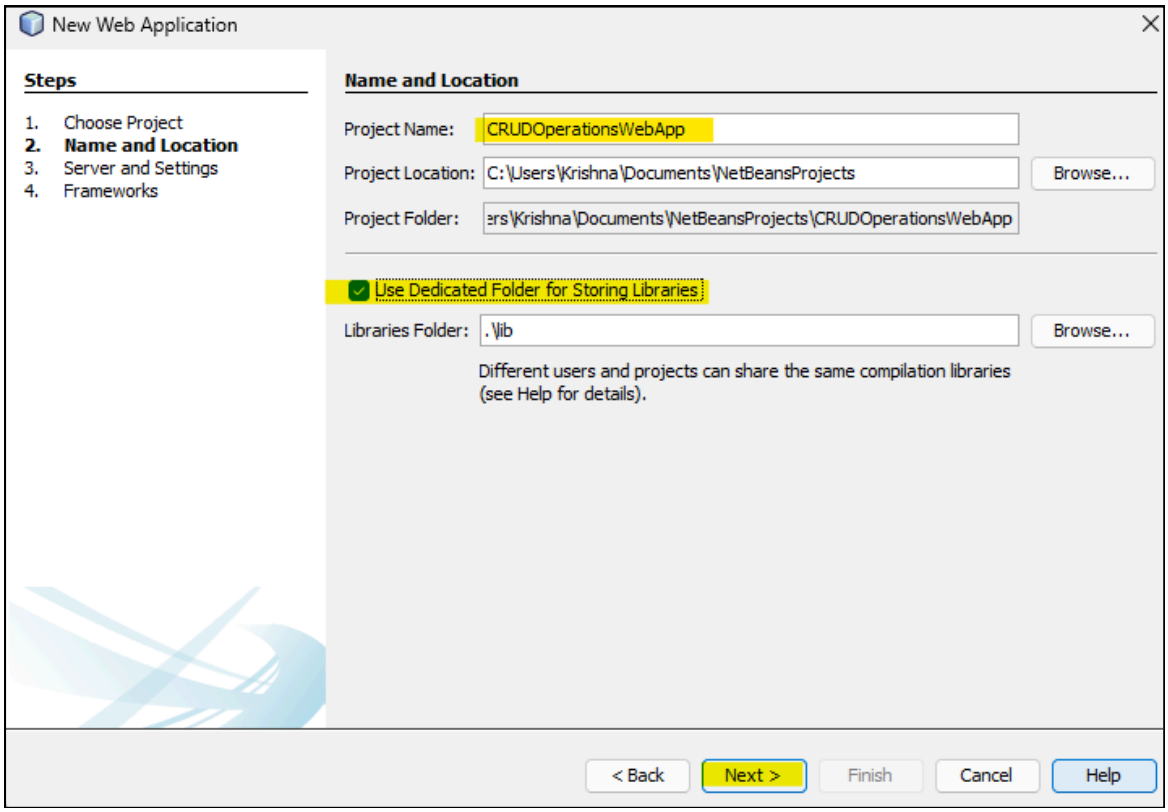

Open NetBeans 8.2, click on File on the left side, and select New Project from the drop-down list.



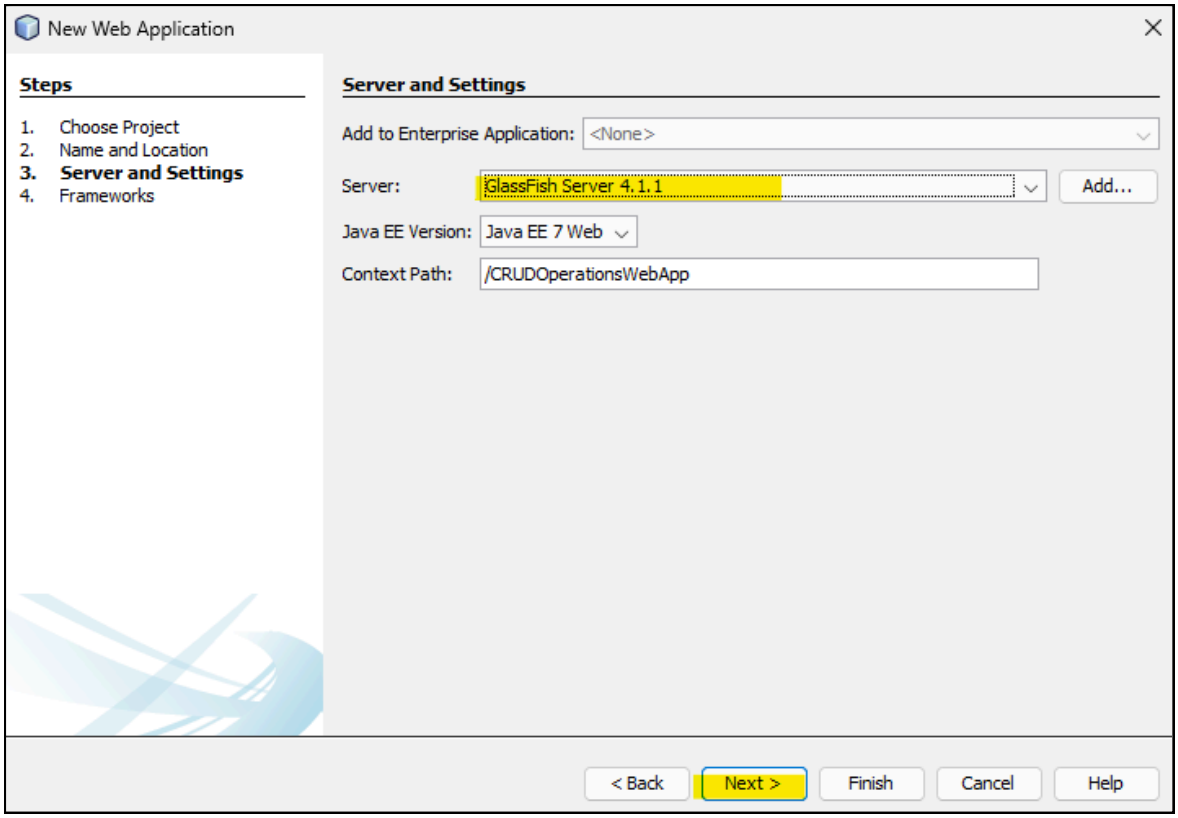
In Categories, select Java Web, then in Projects, select Web Application, and click on the Next button.



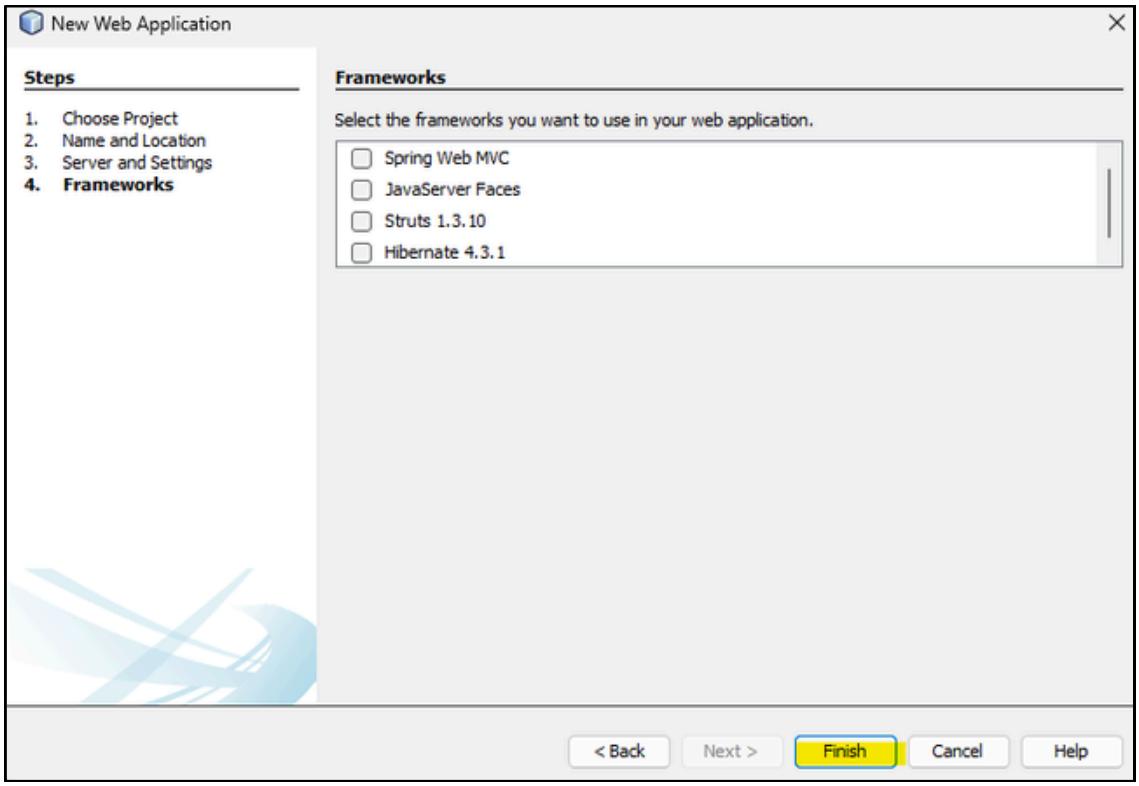
Now create a web application with the name CRUDOperationsWebApp. A window will open like following pic.



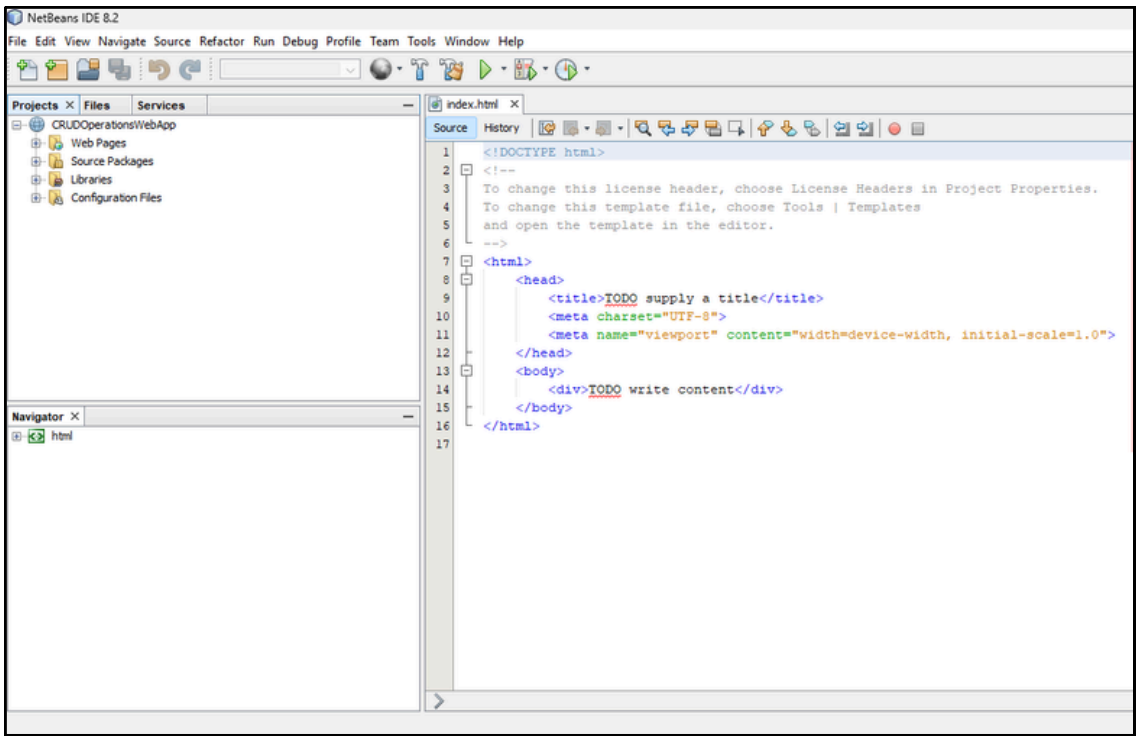
Select GlassFish Server 4.1.1 as the server, and then click on the Next button.



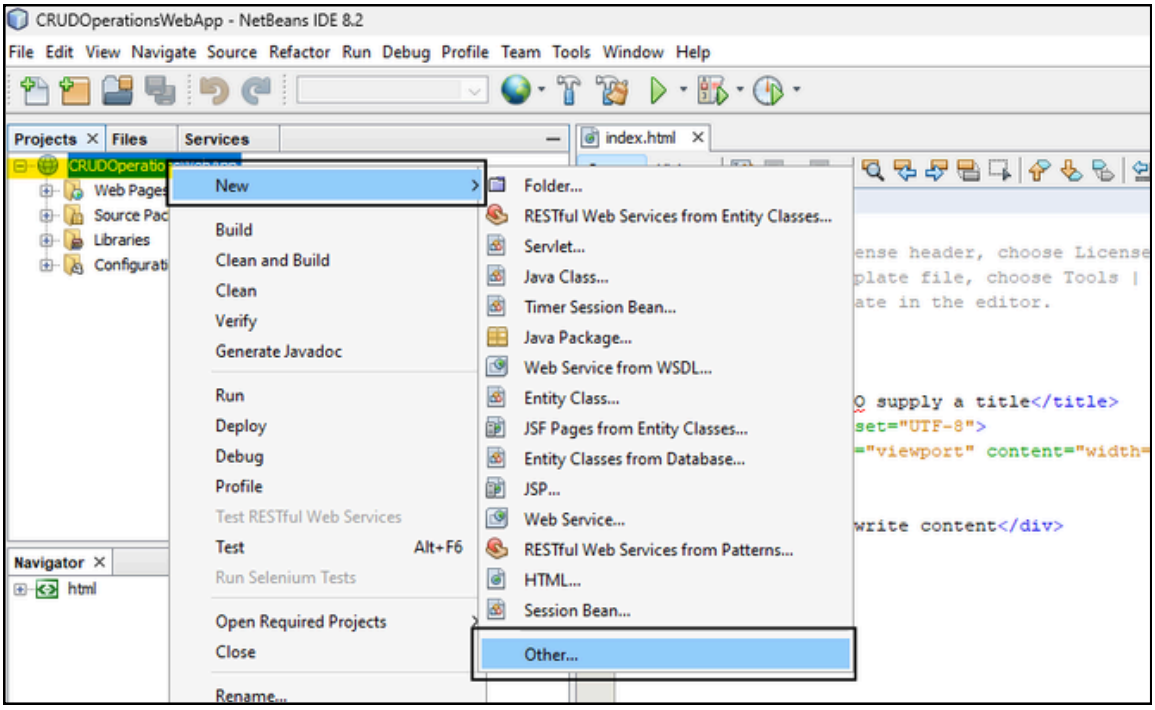
Keep the default settings and click on the Finish button.



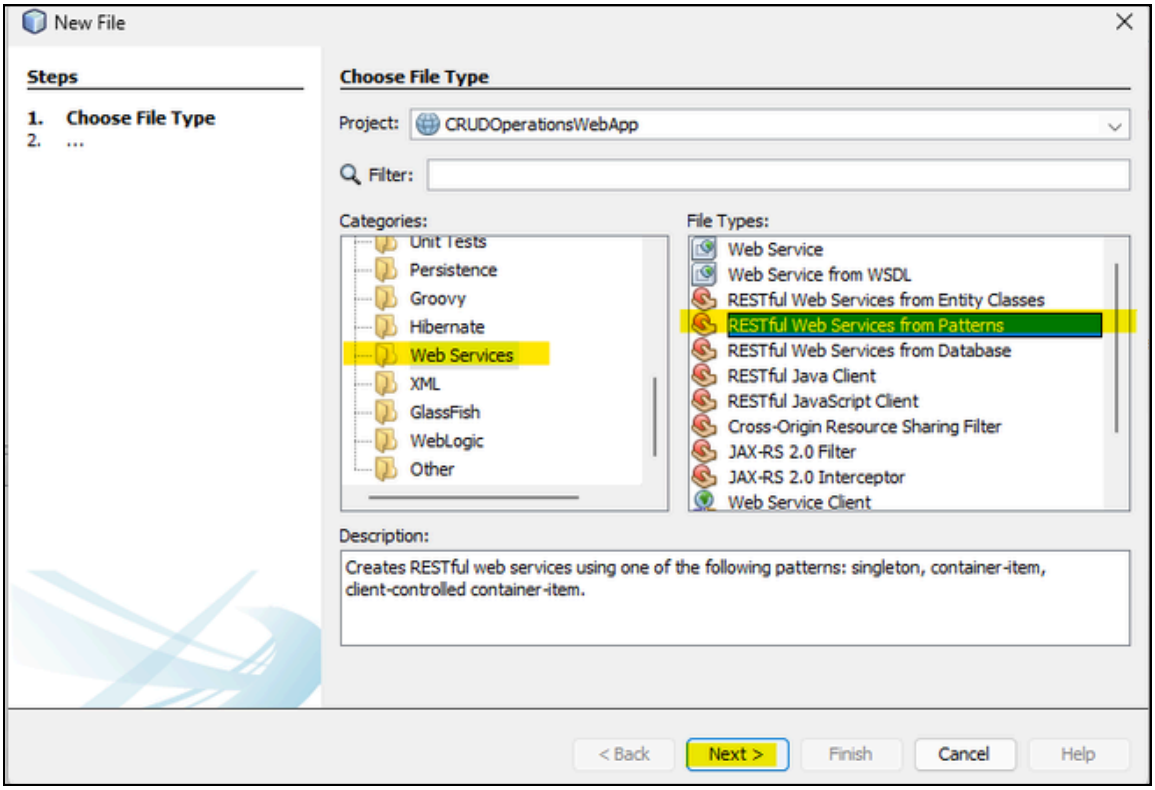
It will look like this.



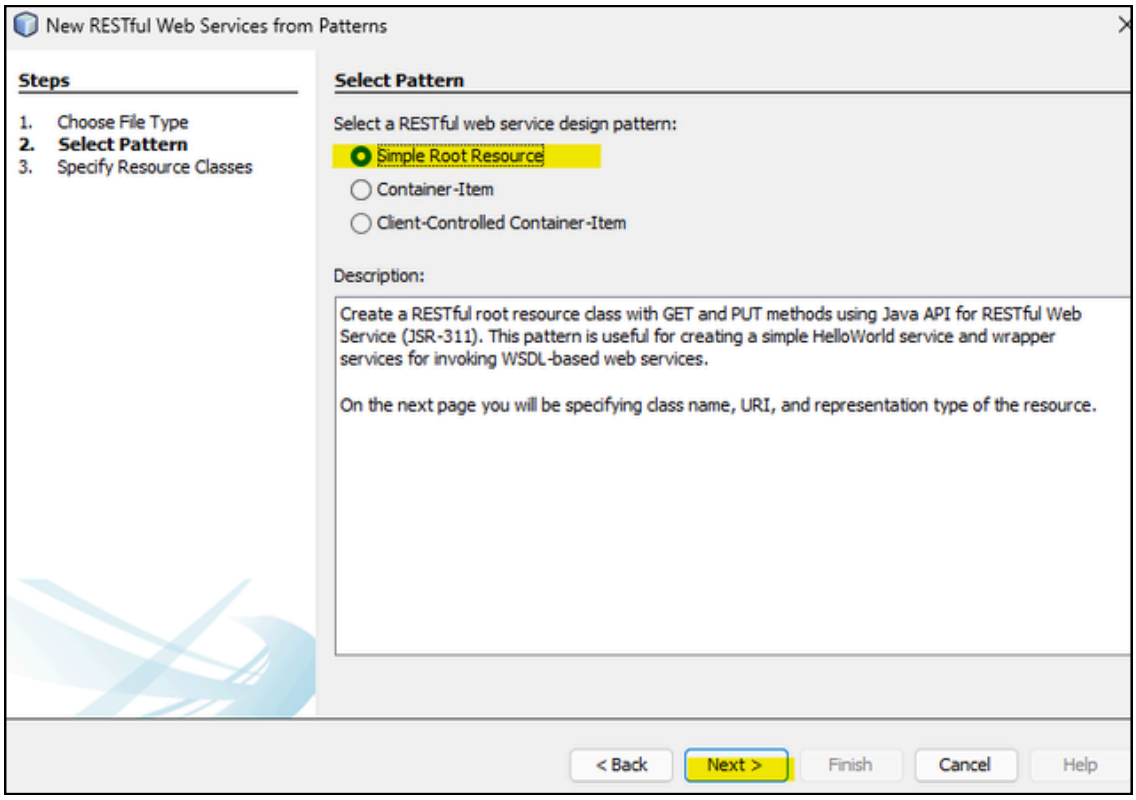
Right-click on CRUDOperationsWebApp, click on New, and then select Other.



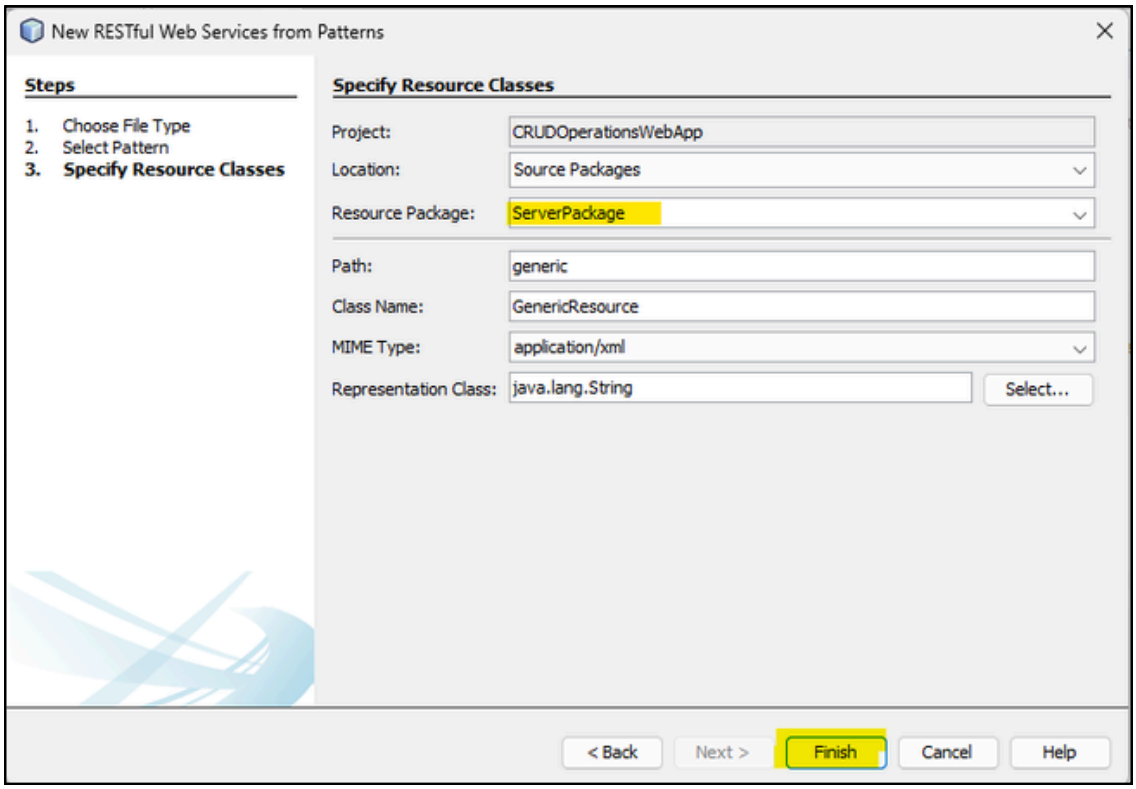
In the categories, select Web Services; in File Types, select RESTful Web Services from Patterns, and then click on the Next button.



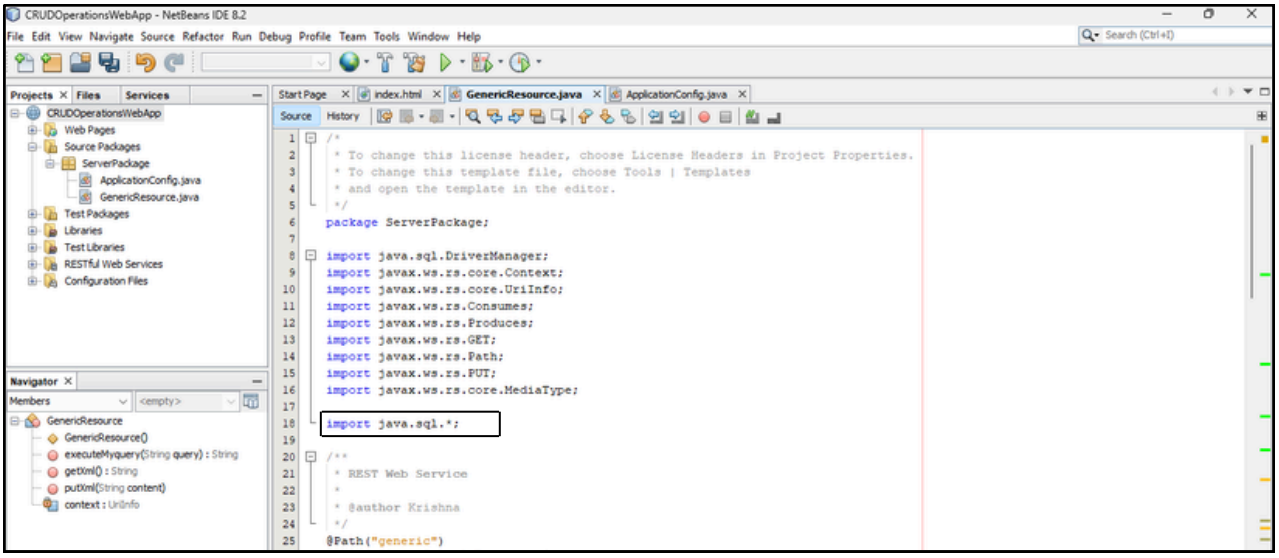
Select Simple Root Resource, and then click on the Next button.



In the Resource Package, enter ServerPackage, keep the rest of the settings as default, and then click on Finish.



Click on GenericResource.java and make the following changes by adding the import statement " **import java.sql.*;** " as shown in the rectangular border.



Add the following code at the end of the original code.

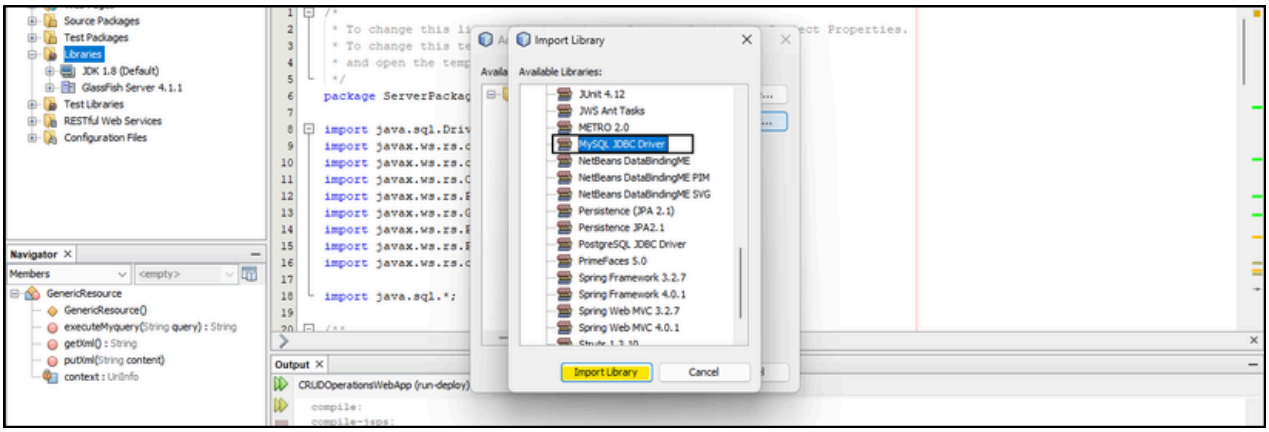
```
@PUT
  @Consumes(MediaType.APPLICATION_XML)
  @Path("/ExecuteQuery")
  public String executeMyquery(String query)
  {
     try
     {
        Class.forName("com.mysql.jdbc.Driver");
        String url = "jdbc:mysql://localhost/prac1";
        Connection con = DriverManager.getConnection(url,"root","root");
        Statement st = con.createStatement();
        String q = query;
        st.execute(q);

        st.close();
        con.close();

        return "Queryis Executed!!!";
     }
     catch (Exception e)
     {
       return e.getMessage();
     }
  }
}
```

```
    @Consumes(MediaType.APPLICATION_XML)
    public void putXml(String content) {
    }

    @PUT
    @Consumes(MediaType.APPLICATION_XML)
    @Path("/ExecuteQuery")
    public String executeMyquery(String query)
    {
        try
        {
            Class.forName("com.mysql.jdbc.Driver");
            String url = "jdbc:mysql://localhost/pracl";
            Connection con = DriverManager.getConnection(url,"root","root");
            Statement st = con.createStatement();
            String q = query;
            st.execute(q);

            st.close();
            con.close();

            return "Queryis Executed!!!";
        }
        catch (Exception e)
        {
            return e.getMessage();
        }
    }
}
```

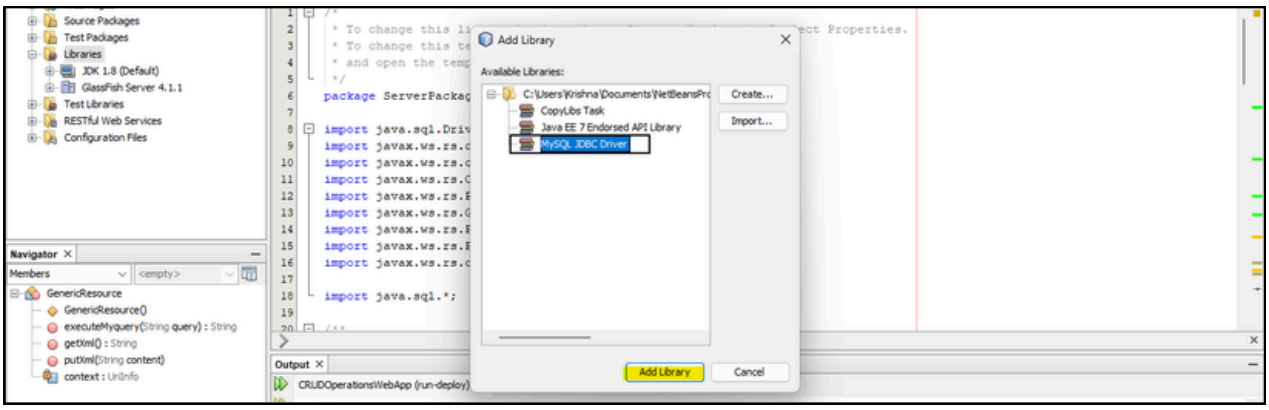
Inside CRUDOperationsWebApp, right-click on Libraries, then click on Add Library.

Click on Import

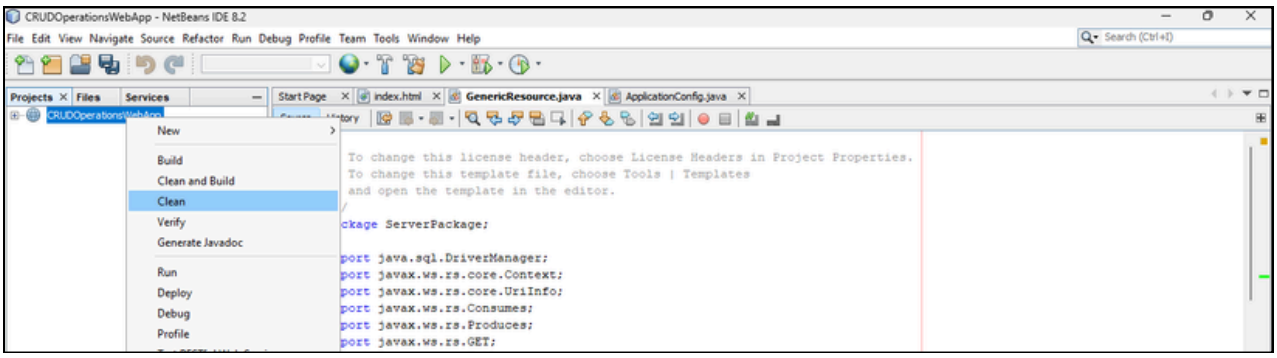
In the Available Libraries list, select MySQL JDBC Driver and click Import Library.

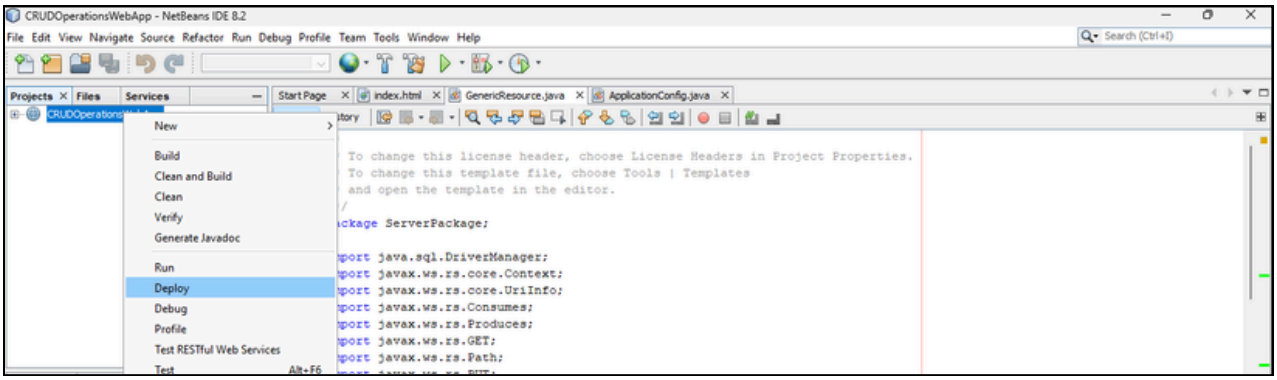


Click on Add library

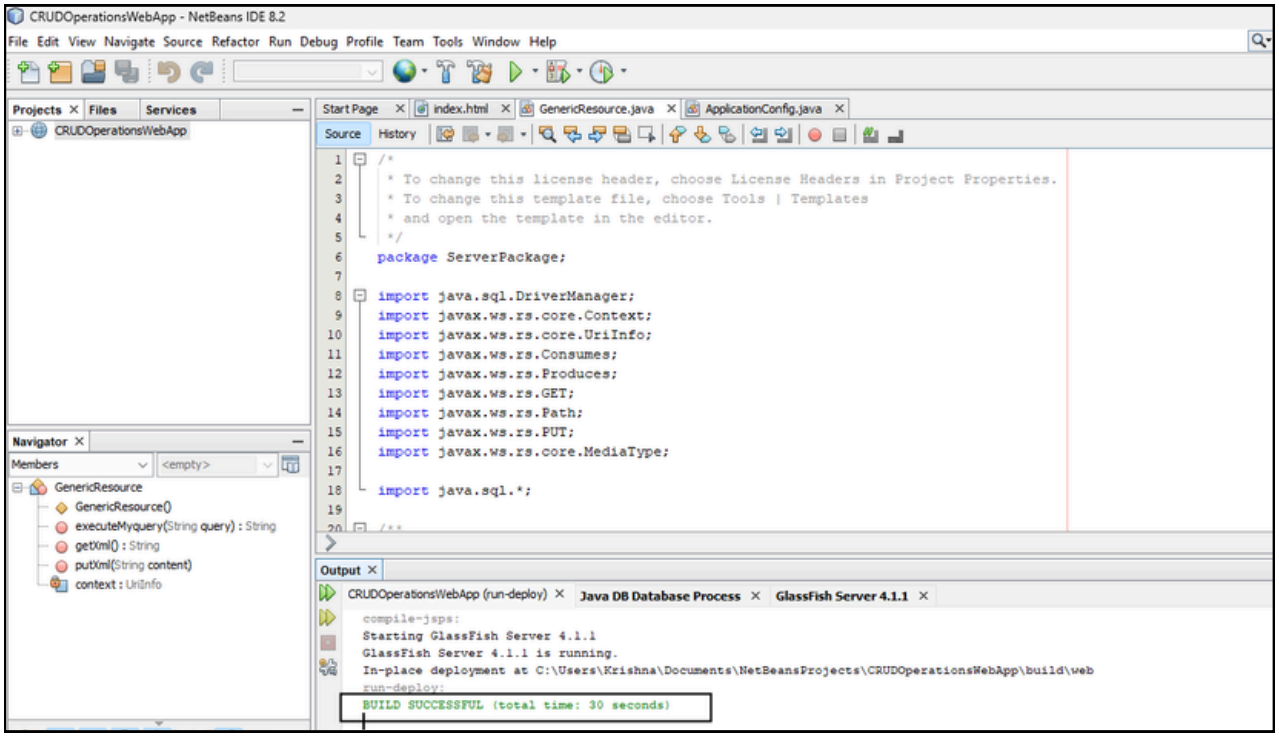


Right-click on CRUDOperationsWebApp, click on Clean, and then deploy the project, as shown in the picture below.

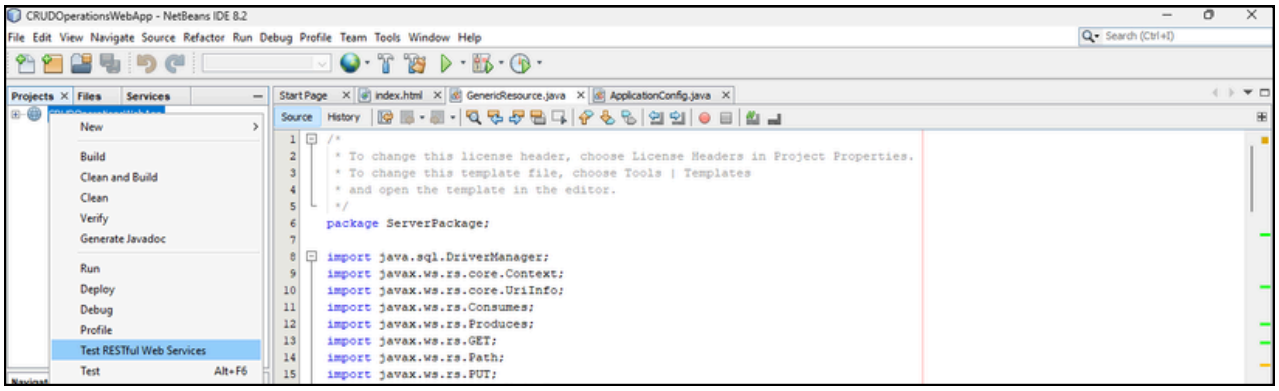


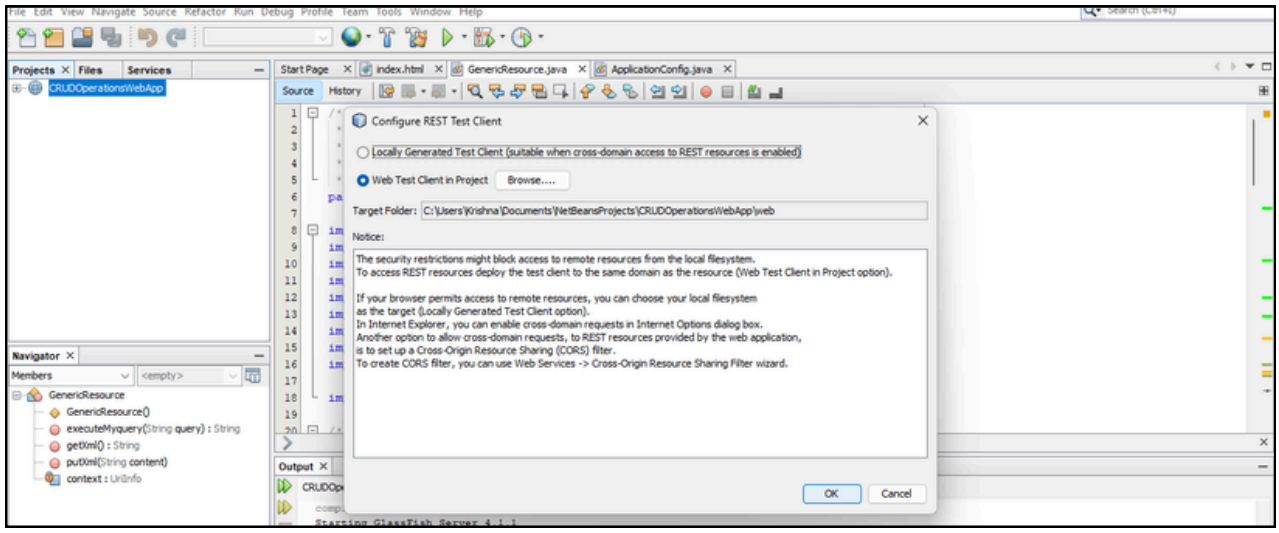


It will display Build Successful at the bottom in the Output section.

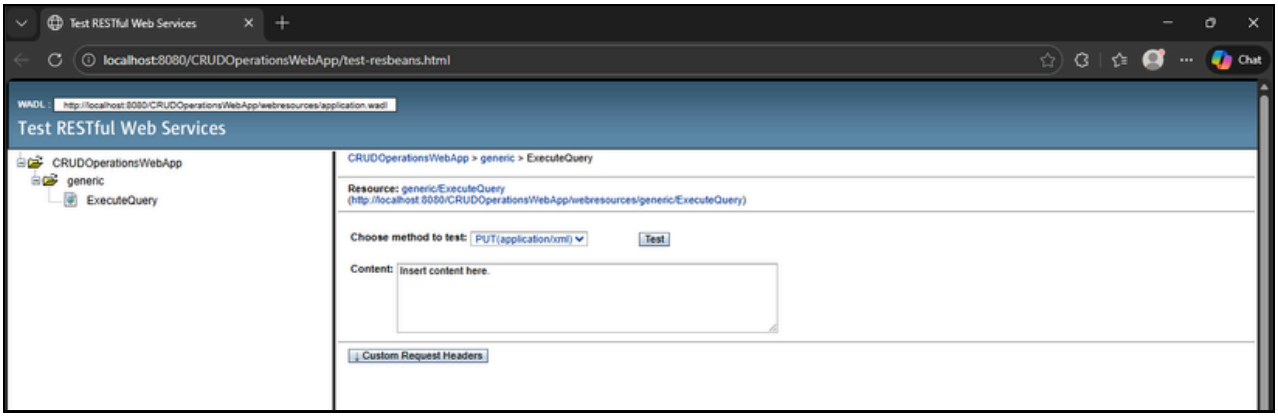


Right-click on CRUDOperationsWebApp, click on Test RESTful Web Services, keep all settings as default, and then click OK.





It will automatically redirect to the browser, and it will look like this.

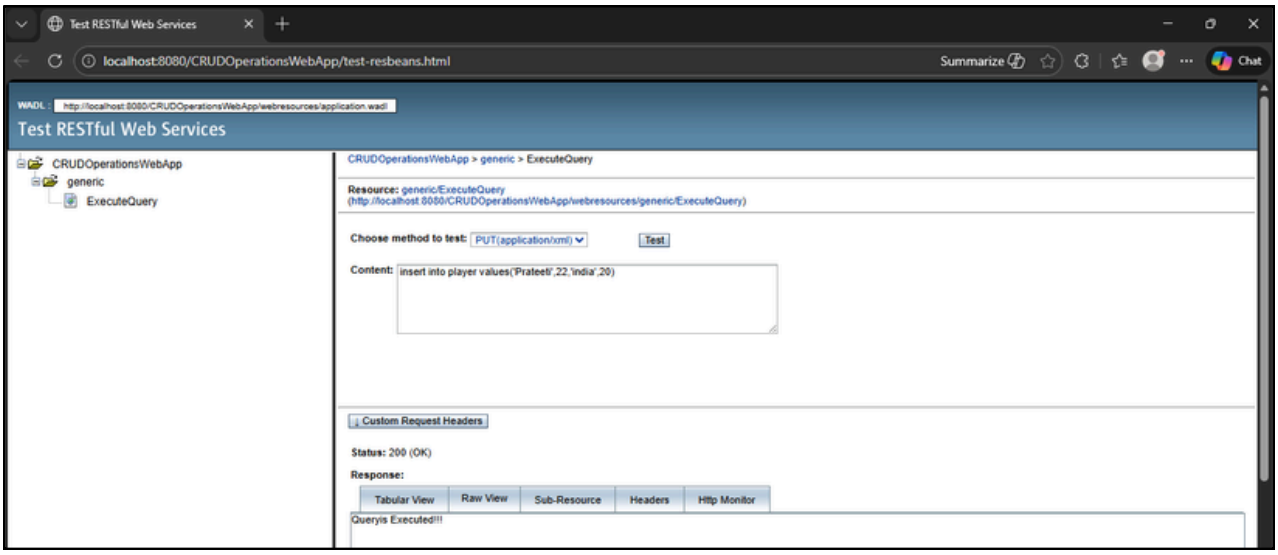


1. **Insert Operation**

   In the Content field, enter the following query:

   insert into player values('Prateeti', 22, 'India', 20)

   Then, click on the Test button. It will display Query Executed!!!.



Repeat the same steps to add two more players, as shown below.

insert into player values('geeta',22,'india',20)

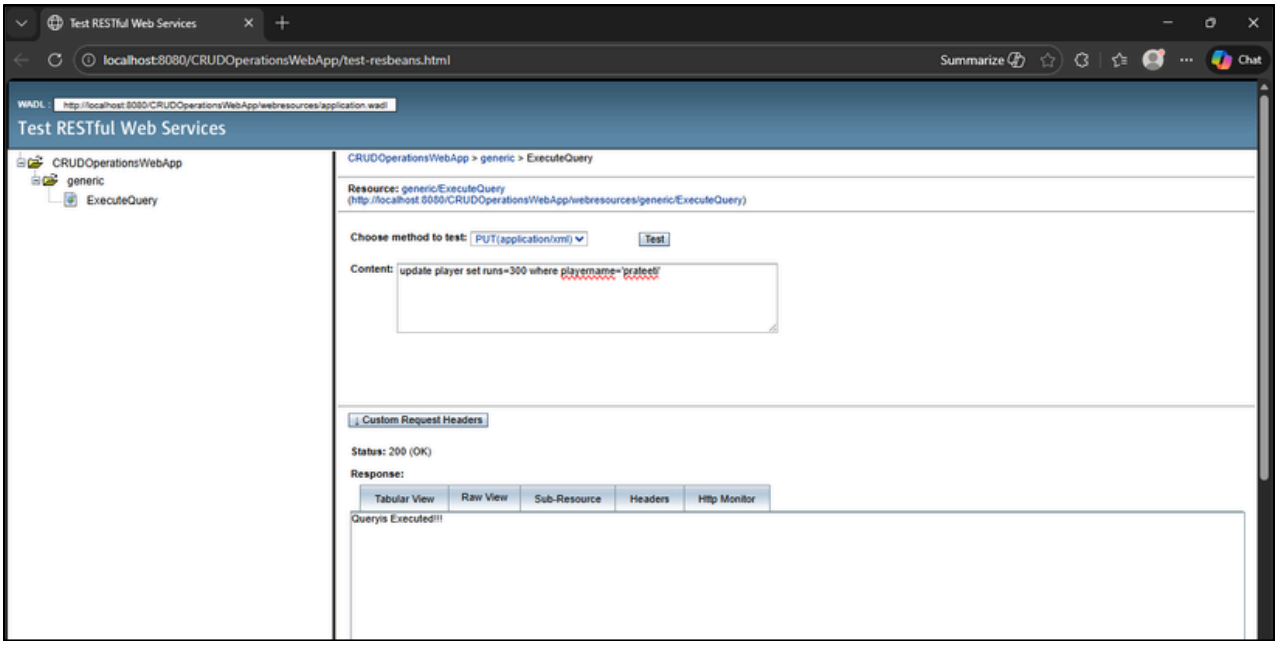
insert into player values('test',22,'india',20)

```
+------------+-----------+---------+------+
| playername | playerage | country | runs |
+------------+-----------+---------+------+
| Prateeti   |        22 | india   |   20 |
| Prateeti   |        22 | india   |  300 |
| geeta      |        22 | india   |   20 |
| test       |        22 | india   |   20 |
+------------+-----------+---------+------+
```

2.Updtae Operation

Now, update the record of Prateeti by changing the runs from 20 to 300 using the following query:

update player set runs = 300 where playername = 'Prateeti';

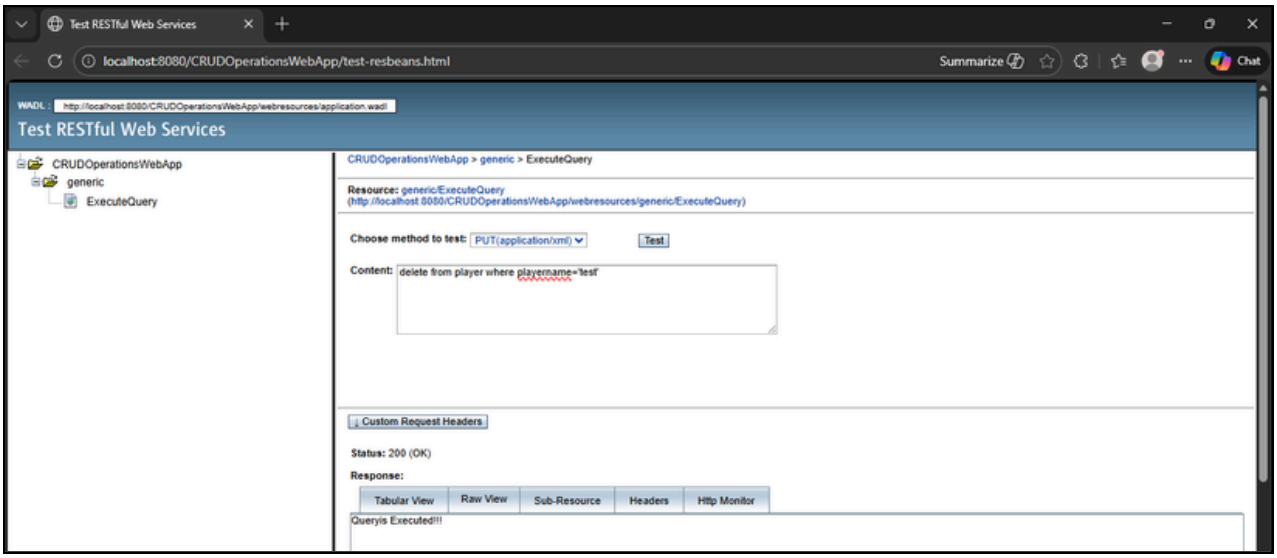


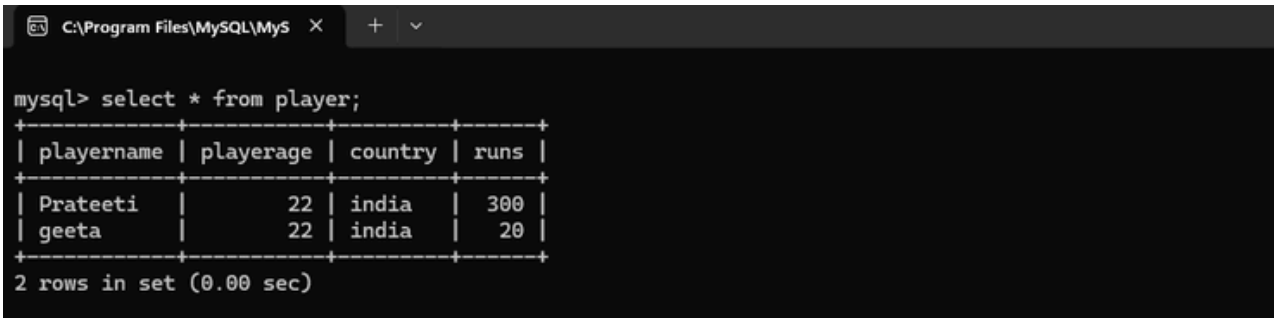

```
mysql> select * from player;
+------------+-----------+---------+------+
| playername | playerage | country | runs |
+------------+-----------+---------+------+
| Prateeti   |        22 | india   |  300 |
| geeta      |        22 | india   |   20 |
| test       |        22 | india   |   20 |
+------------+-----------+---------+------+
```

3.Delete Operation

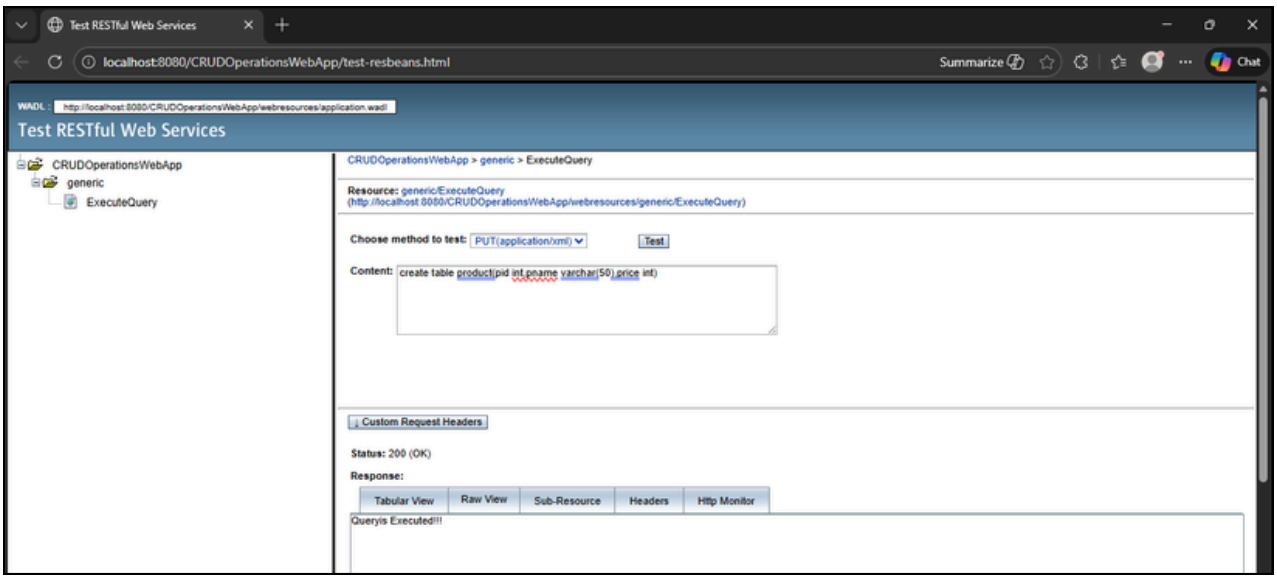
Now, delete the record of test using the following query:

delete from player where playername = 'test';





4.Creating table

Create the product table using the following query:

create table product (pid int, pname varchar(50), price int);